Energy Resilience Under Prolonged Grid Outages: Which Source Holds When the Grid Doesn’t?

Case study: ATÜ campus, Adana · Ahmet Selahaddin Topbaş · TOPBASCH Systems Lab · Planning stage, no results yet

Purpose. This register lists every source consulted so far, with the depth to which each was read and what was taken from it. Reading depth is stated honestly: *Read (relevant sections)* means the sections that bear on this project were read; *Abstract read* means only the abstract (and sometimes the introduction) was read, so nothing beyond it is claimed. Publisher PDFs are not re-hosted; links lead to the publisher or an open-access copy.

Papers and reviews listed	Read (relevant sections)	Abstract read	Search record
28	6	22	Scopus and web, 30 Oct 2026, ten queries (section 6)

1 · Method sources and closest works

The outage-metric names and the every-start-hour scan come from this group. Method and names are adopted; findings are not.

Code	Source	Read depth	What was taken / note
D21	**Wodicker, Nelson, Johnson (2024), Unified dispatch of grid-connected and islanded microgrids, Frontiers in Energy Research 11** https://www.frontiersin.org/articles/10.3389/fenrg.2023.1257050/full	Read (relevant sections)	Definitions of survivability (probability of serving 100% of critical load), autonomy (duration) and unserved energy; the all-start-times scan. Method and names are adopted, not claimed as new. Findings are not used. Military microgrid, diesel + battery + PV, outage started at every hour of the year over a 7-day horizon; emissions reported. No hydrogen, no campus.
D29	**Marqusee, Becker, Ericson (2021), Resilience and economics of microgrids with PV, battery storage, and networked diesel generators, Advances in Applied Energy 3, 100049** https://doi.org/10.1016/j.adapen.2021.100049	Read (relevant sections)	Critical-load coverage probability by outage start (t) and duration (d); generator and battery reliability; campus-like critical load. Method adopted, not claimed as new. No hydrogen, no earthquake region. 114 citing documents in Scopus (30 Oct 2026); only one of them (Lee 2026) combines hydrogen with outage survivability.

Code	Source	Read depth	What was taken / note
D30	**Nelson, Johnson et al. (2020), Statistical development of microgrid resilience during islanding operations, Applied Energy** https://doi.org/10.1016/j.apenergy.2020.115724	Abstract read	Survivability = probability of meeting critical load during an islanding event. Military microgrid; Markov chain over a 7-day horizon.
D31	**Peterson, Van Bossuyt et al. (2021), Analyzing mission impact of military installations microgrid for resilience, Systems 9(3), 69** https://doi.org/10.3390/systems9030069	Abstract read	Resilience expressed as impact on mission objectives rather than cost.
D27	**Lee (2026), Design-actionable cost-resilience thresholds for islanded battery-hydrogen microgrids, Int. J. Hydrogen Energy 264, 156805** https://doi.org/10.1016/j.ijhydene.2026.156805	Read (relevant sections)	8760-hour dispatch; four random outage starts per year, fixed 48 h, worst case over 10 seeds; battery-hydrogen holds 48 h while a diesel-battery system loses survivability under derating, fuel limits or start-up delay. Differences from this project: fixed duration, random starts, off-grid island, example load. Island microgrid (Philippines), calibrated example load; cited by 0 documents in Scopus on 30 Oct 2026. Code and data are open (GitHub, Zenodo; see the paper).
D28	**Dong et al. (2018), Battery and backup generator sizing for a resilient microgrid under stochastic extreme events, IET Generation, Transmission & Distribution 12(20)** https://doi.org/10.1049/iet-gtd.2018.5883	Abstract and introduction read	Outage time and duration are treated as uncertain in sizing; supports the "duration is unknown" reasoning. A sizing problem, not an endurance map. Grid-tied PV microgrid; battery and generator sizing (MILP); no hydrogen.

2 · Neighbouring works (outage resilience with hydrogen, campuses, regret, heat waves)

Code	Source	Read depth	What was taken / note
D32	**Kerkau, Fujita, Sepasi, Roose (2026), Resiliency-centric optimization for microgrid sizing with renewable and hydrogen energy infrastructure, IEEE Access** https://doi.org/10.1109/ACCESS.2026.3704950	Abstract read	Outage scenarios of three days with 50% and 100% critical-load support. One duration, a sizing problem. Hawaii test site; MILP + PSO sizing.

Code	Source	Read depth	What was taken / note
D33	**Sharifpour, Ameli, Ameli, Strbac (2023), A resilience-oriented approach for microgrid energy management with hydrogen integration during extreme events, Energies 16(24), 8099** https://doi.org/10.3390/en16248099	Abstract read	Critical-load supply of 82% to 100% across disaster scenarios of different outage durations. Optimisation, not a source-comparison map. Test system of three microgrids; preventive scheduling.
D34	**Dhankar, Chen, Tong (2024), Enhancing microgrid resilience with green hydrogen storage, IEEE PES General Meeting** https://doi.org/10.1109/PESGM51994.2024.10689148	Abstract read	Uses loss of load, duration of outage and system cost as measures; control focus. Model predictive control of electrolyzer and fuel cell.
D35	**Feng et al. (2026), A bi-level resilience-oriented planning framework for hydrogen-integrated multi-energy systems under extreme events, Applied Energy 426, 128671** https://doi.org/10.1016/j.apenergy.2026.128671	Abstract read	Critical-load continuity constraints; planning, not endurance of an existing system. Planning and pricing; Shapley attribution.
D36	**Narayanan, Al-Greer et al. (2026), A priority-aware resilience assessment framework for PV-battery campus microgrids: University of Mosul case study** https://doi.org/10.1109/ICMISI69868.2026.11584107	Abstract read	Tiered campus load (critical, essential, non-essential) is not new; cited as a neighbour. 24-hour simulation; no generator, no hydrogen.
D37	**Kajal et al. (2025), Techno-economic and resilience optimization of a university microgrid: a dual-tool approach, Energy Reports** https://doi.org/10.1016/j.egyr.2025.10.034	Abstract read (full text to be read)	Appears cost and sizing oriented; whether it scans outage duration is to be checked in the full text.
D38	**Yazdanie et al. (2025), Strengthening energy system resilience planning under uncertainty by minimizing regret, Renewable and Sustainable Energy Transition** https://doi.org/10.1016/j.rset.2024.100093	Abstract read	Regret is an accepted method here; this project applies it to unknown outage duration and source choice. Case: Accra; unplanned outages are one of the uncertainty scenarios.
D43	**Şekerci et al. (2026), Sustainable and resilient hydrogen supply chain planning under uncertainty: Marmara Region, Türkiye, Sustainability 18(12), 6112** https://doi.org/10.3390/su18126112	Abstract read	Regional planning (MILP); no campus, no hourly outage scan.
D46	**Islam et al. (2022), Outage survivability investigation of a PV/battery/CHP system in a hospital building in Texas, Sustainability 14(22), 14965** https://doi.org/10.3390/su142214965	Abstract read	No hydrogen.

Code	Source	Read depth	What was taken / note
D47	**Ullah et al. (2025), Cost-aware strategies for enhancing energy resilience in microgrids via stationary and mobile resources, J. Safety Science and Resilience** https://doi.org/10.1016/j.jnlssr.2024.10.002	Abstract read	No hydrogen.
D48	**Sheng et al. (2023), Assessing thermal resilience of an assisted living facility during heat waves and cold snaps with power outages, Building and Environment** https://doi.org/10.1016/j.buildenv.2023.110001	Abstract read	Buildings, not source comparison.

Earlier-stage readings, kept in the register but not cited in the Blueprint

Code	Source	Read depth	What was taken / note
D22	**Atef et al. (2026), Feasibility-aware energy management of a hydrogen-enabled community microgrid (arXiv preprint, not peer-reviewed)** https://arxiv.org/pdf/2607.17248	Abstract read	Outages modelled as single isolated hours; the authors state that sustained multi-day outages were not tested. Preprint; community in Queensland, not a campus. Not cited in Blueprint v2.0; kept as background.
D23	**Gergin, Alemin, Aksan (2026), Hybrid renewable energy systems for post-disaster settlements: Cinarcik (engrXiv preprint, not peer-reviewed)** https://engrxiv.org/preprint/view/6181	Abstract read	Design and cost study (HOMER Pro); outage-duration treatment not seen in what I read.
D24	**Ale et al. (2026), Can distributed hydrogen production improve the earthquake resilience of power systems? (Applied Energy 403, journal to be confirmed)** https://investigadores.anid.cl//en/public_search/work?id=839025	Abstract read	New Zealand national-scale planning, not a campus.
D25	**Karaman et al. (2023), Enhancing resiliency of integrated space-air-ground-sea networks with renewable energies: after the 2023 Türkiye earthquake (arXiv preprint)** https://arxiv.org/pdf/2405.17635	Abstract read	About telecom base stations, not campuses; the generator figures are the authors' account, to be checked at the source.
D26	**Lin, Zhang, Lu (2023), Resilience-assuring hydrogen-powered microgrids (arXiv preprint)** https://arxiv.org/pdf/2312.04014	Abstract read	6-hour window on a test feeder; no real site.

3 · Reviews used to test the gap statement

Code	Source	Read depth	What was taken / note
D39	**Yu et al. (2026), A review of hydrogen-enabled resilience enhancement for multienergy systems, Proceedings of the IEEE 114(5)** https://doi.org/10.1109/JPROC.2026.3711369	Read (gaps section and outage models)	Two outage models exist (fixed-length islanding, stochastic outage). Gaps: hydrogen-specific metrics (availability, pressure; load loss and carbon jointly) and scenarios of prolonged heat waves. Tank pressure is outside this project's model. Does not list this project's combination as a gap; it also does not contradict it.
D40	**She et al. (2026), A review of energy storage for power system resilience: functions, metrics, and applications, Applied Energy** https://doi.org/10.1016/j.apenergy.2026.128056	Read (metrics section)	Resilience metrics lack standardisation; adequacy metrics include expected unserved energy. Survivability and autonomy were not seen in the sections read.

4 · Earthquake and Türkiye context

No official duration of the February 2023 power outage in Adana or at the campus was found (D15 remains open). No outage duration is assumed in the model; all durations are scanned.

Code	Source	Read depth	What was taken / note
D41	**Akbaş et al. (2025), Enhancing resilience with mobile substations: lessons from TEİAŞ's response to the Kahramanmaraş earthquake, CIGRE Science & Engineering 39**	Read (section 3)	Mobile substations energised within days; in Hatay and Gaziantep the damaged distribution network carried only 3 to 5 MVA over several days; mobile coverage was unavailable for days in some zones. No duration for Adana or the campus. Not used as a number.
D42	**Ozen, Kaya (2023), Earthquake effects on electricity network: a case study in Turkish grid, IEEE GPECOM 2023** https://doi.org/10.1109/GPECOM58364.2023.10175728	Abstract read	Duration not stated in the abstract; full text to be checked.
D07	**Official Gazette 10 Feb 2023, no. 32100: state-of-emergency decision no. 1354**	Read	–
D08	**Ministry of Environment, Urbanisation and Climate Change statement, 16 Feb 2023**	Read	Other media figures not yet cross-checked with official bodies
D15	**Official duration of the 2023 power outage in Adana (EPDK, TEİAŞ, EDAŞ, AFAD)**	Searching	No official figure found so far
D50	**NREL ESIF 2021 annual report (docs.nrel.gov/fy21osti/78837)** https://docs.nrel.gov/docs/fy21osti/78837.pdf	Read (not relevant)	Nothing relevant to the question. Correction: my earlier guess about its content was wrong. The paper I was looking for is D29.

5 · Japan side and planned sources

Code	Source	Read depth	What was taken / note
D44	**Marnay, Aki, Hirose et al. (2015), Japan's pivot to resilience: how two microgrids fared after the 2011 earthquake, IEEE Power & Energy Magazine** https://doi.org/10.1109/MPE.2015.2397333	Abstract read	–
D45	**Aki, Sugimoto, Ishida (2014), A concept of hydrogen energy community to improve demand side energy resiliency, 20th World Hydrogen Energy Conference**	Abstract read	Load priorities matter for maximising resilience, in line with the tier idea here. Concept paper; no numerical scan.
D11	**Kashiwa hourly weather data (JMA)**	Planned (W02)	–
D12	**Japan Ministry of the Environment electricity emission factors**	Planned (W02)	Values to be checked at the source

6 · Search record (Scopus and web, 30 October 2026)

Query	Theme	Records	Outcome
T2-S1	Microgrid outage resilience and hydrogen	174	Too broad; replaced by S1b
T2-S1b	Narrowed version of S1	14	Found Lee (2026), the closest work
T2-S2, S2b, S2c	Strict combination of terms, then relaxed versions	0, 7, 4	S2 too restrictive; S2b and S2c reviewed
T2-S3 to S9	Campus microgrids, survivability, regret, heat waves, Türkiye, Japan	Exported and screened	Neighbouring works in sections 1 to 5
T2-S10a to c	Papers citing Marqusee 2021, Wodicker 2024, Lee 2026	114, 11, 0	Of 114 citing Marqusee: hydrogen or fuel cell 55; outage or islanding 45; both 19. Only Lee (2026) combines hydrogen with outage survivability.

Finding, worded as a limited search. In a search limited to Scopus and the web on 30 October 2026, no work offering the combination of four sources, one outage scan along a continuous duration axis, heat-wave and refuelling conditions, a regret table and a campus case with a Kashiwa comparison was seen. This is not proof that none exists. The closest work is Lee (2026): fixed 48 h outages, random start times, island microgrid, calibrated example load.

Limits. Scopus and web only; titles and abstracts were screened for most records. Works outside these sources, in other languages or not yet indexed would not be seen. The register is re-read at the first-month review.

Corrections on record. (1) A file expected to be a microgrid resilience paper was a facility annual report (D50); the intended paper is Marqusee et al. (D29). (2) A wrong link for the CIGRE article was removed; it is cited by issue (D41).

7 · Programme and site sources

Code	Source	Read depth	Note
D01	**UTSIP official application page (2026 cycle)**	Read	Application route and the number of labs that can be chosen.
D02	**UTSIP 2026 host lab list**	Read	The 2027 list is expected around the end of November 2026.
D16	**UTSIP past presentations 2024 to 2026**	Read	Used to see how past projects are titled.
D19	**UTSIP leaflet 2024**	Read	Programme overview.
D20	**UTSIP programme page (International Liaison Office)**	Read	Programme overview and eligibility. https://www.ilo.k.u-tokyo.ac.jp/summer_en
D51	**Wikipedia pages on ATÜ, Kashiwa Campus and Adana**	Read	Secondary source for general site facts; to be replaced by official pages.